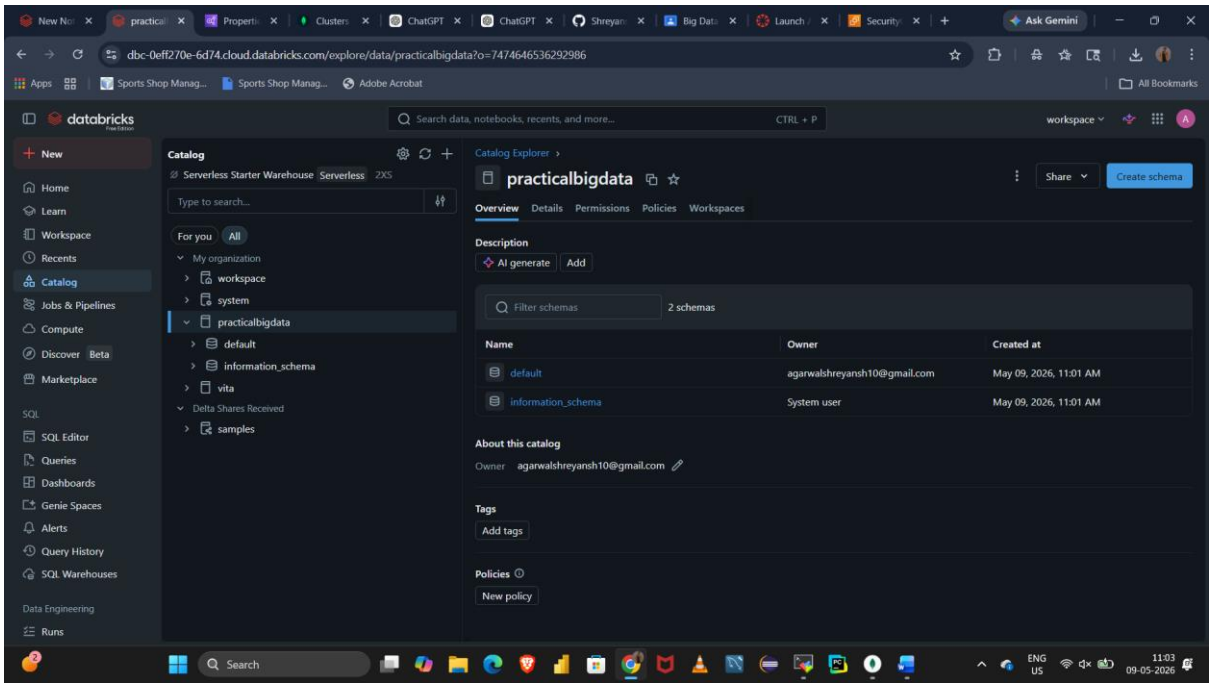
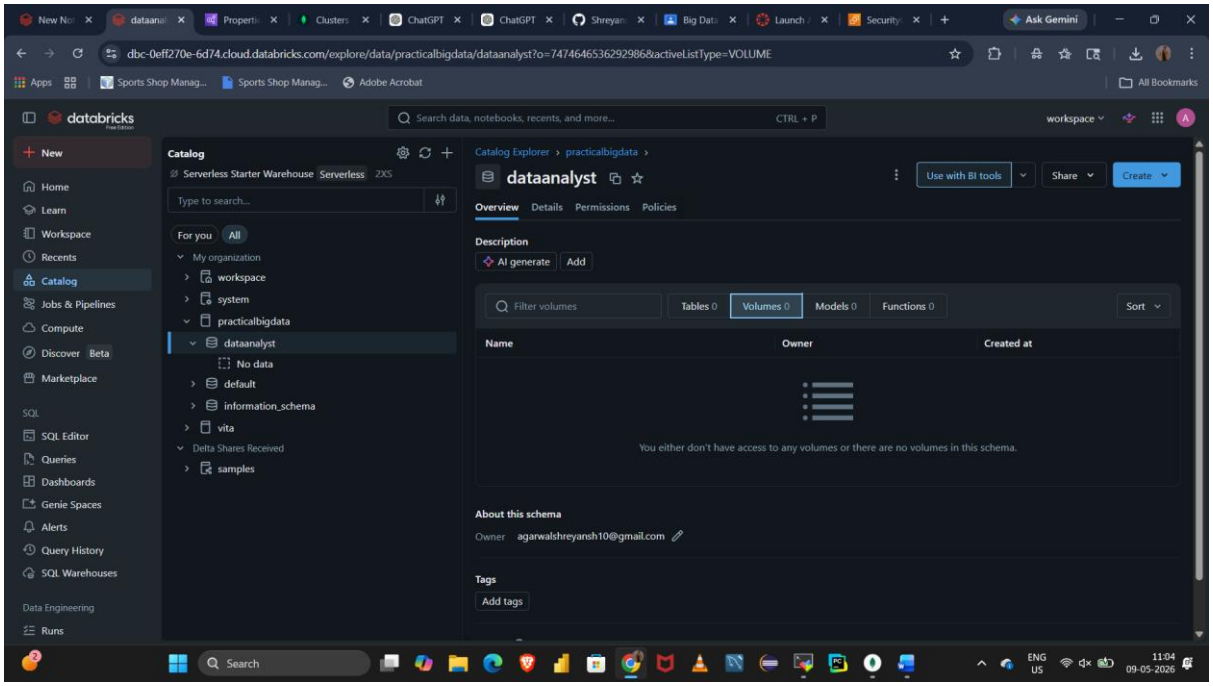
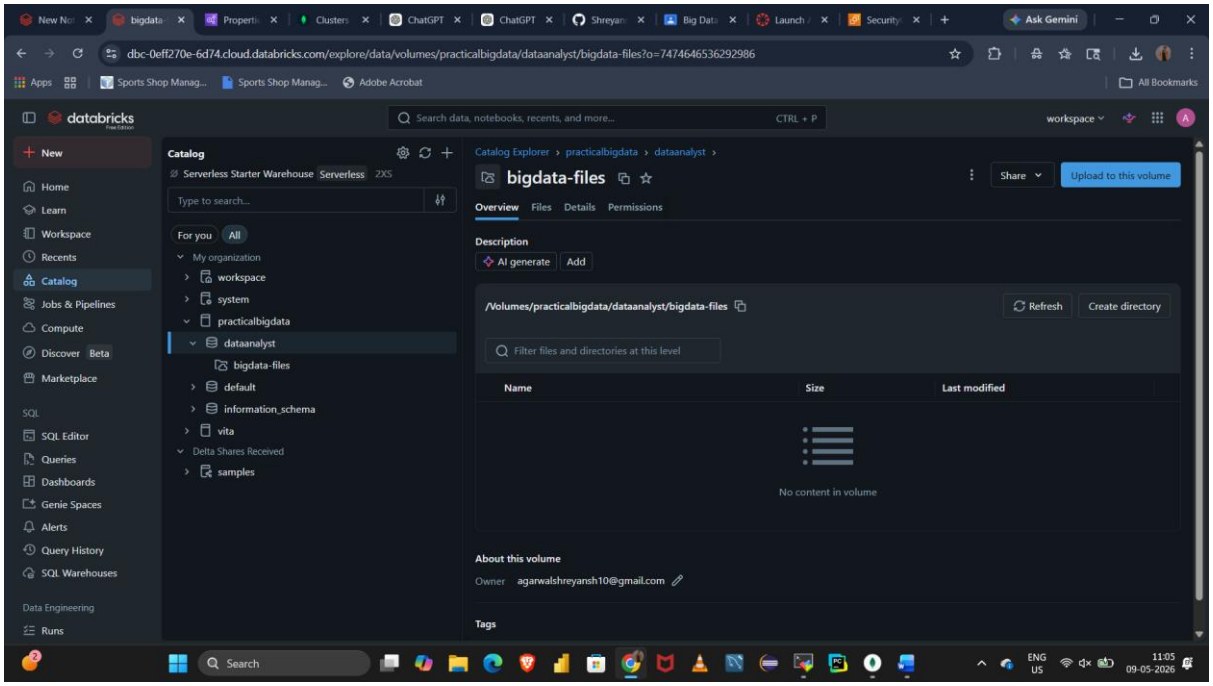
New catalog



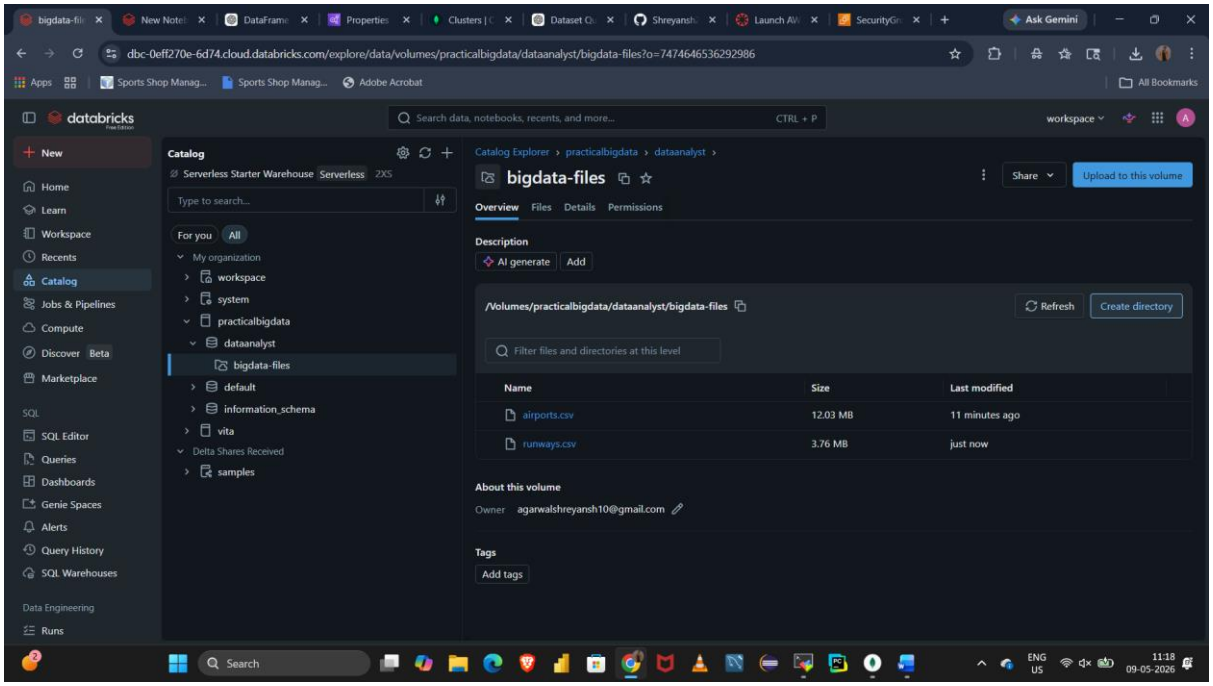
New schema



New volume



Insert files in volume



Queries

1.

dbc-0eff270e-6d74.cloud.databricks.com/editor/notebooks/1528599417974162?o=7474646536292986#command/7311693347752004

New Notebook 2026-05-09 10:58:58

Just now (10s) 1

```
df_airports = spark.read.csv("/Volumes/practicalbigdata/dataanalyst/bigdata-files/airports.csv",header=True,inferSchema=True)
```

df_airports: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 17 more fields]

Just now (1s) 2

```
df_runways = spark.read.csv("/Volumes/practicalbigdata/dataanalyst/bigdata-files/runways.csv",header=True,inferSchema=True)
```

df_runways: pyspark.sql.connect.dataframe.DataFrame = [id: integer, airport_ref: integer ... 18 more fields]

Just now (<1s) 3

```
from pyspark.sql.functions import col, count, avg, sum, max
```

Just now (<1s) 4

```
df_airports = df_airports.withColumn("latitude_deg", col("latitude_deg").cast("double")) \
                         .withColumn("longitude_deg", col("longitude_deg").cast("double")) \
                         .withColumn("elevation_ft", col("elevation_ft").cast("int"))

df_runways = df_runways.withColumn("length_ft", col("length_ft").cast("int")) \
                       .withColumn("width_ft", col("width_ft").cast("int"))
```

dbc-0eff270e-6d74.cloud.databricks.com/editor/notebooks/1528599417974162?o=7474646536292986#command/7311693347752004

New Notebook 2026-05-09 10:58:58

Just now (<1s) 4

```
                         .withColumn("longitude_deg", col("longitude_deg").cast("double")) \
                         .withColumn("elevation_ft", col("elevation_ft").cast("int"))

df_runways = df_runways.withColumn("length_ft", col("length_ft").cast("int")) \
                       .withColumn("width_ft", col("width_ft").cast("int"))
```

df_airports: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 17 more fields]

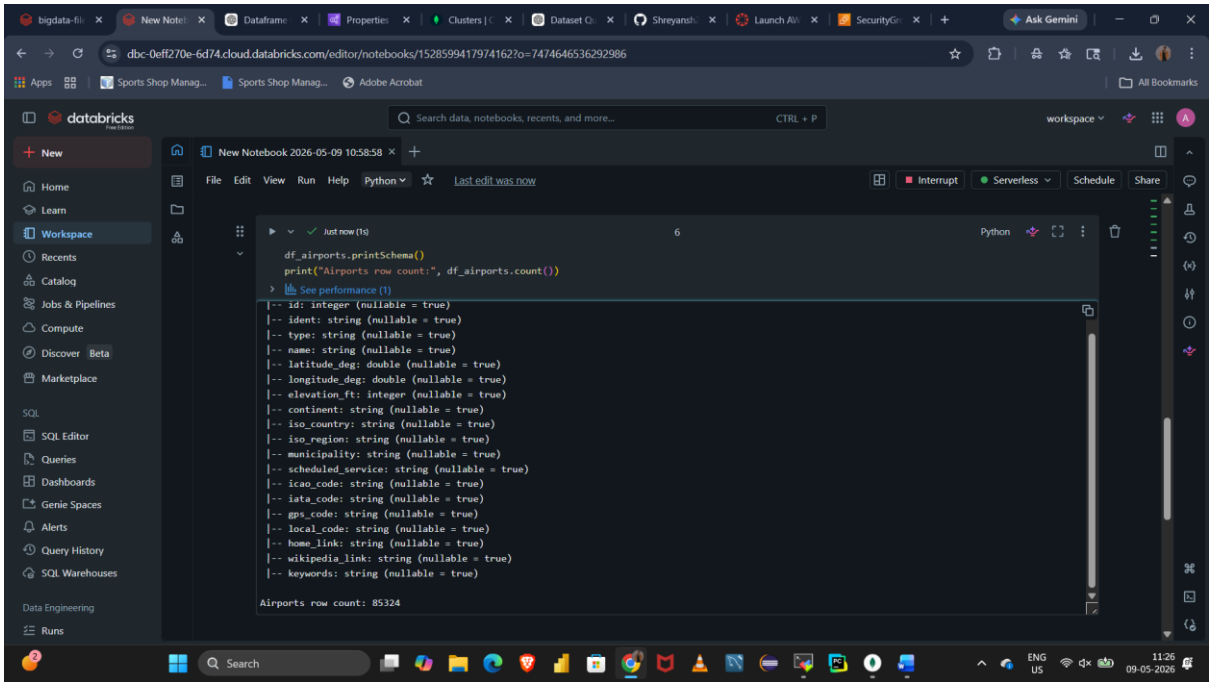
df_runways: pyspark.sql.connect.dataframe.DataFrame = [id: integer, airport_ref: integer ... 18 more fields]

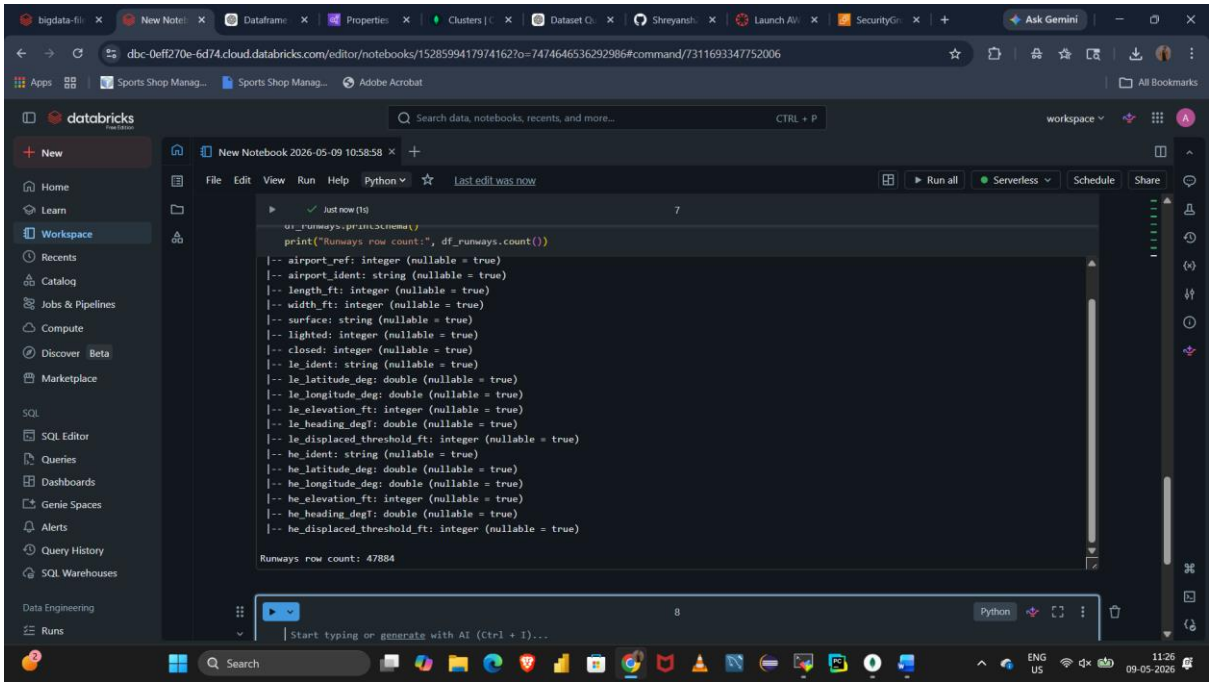
Just now (1s) 5

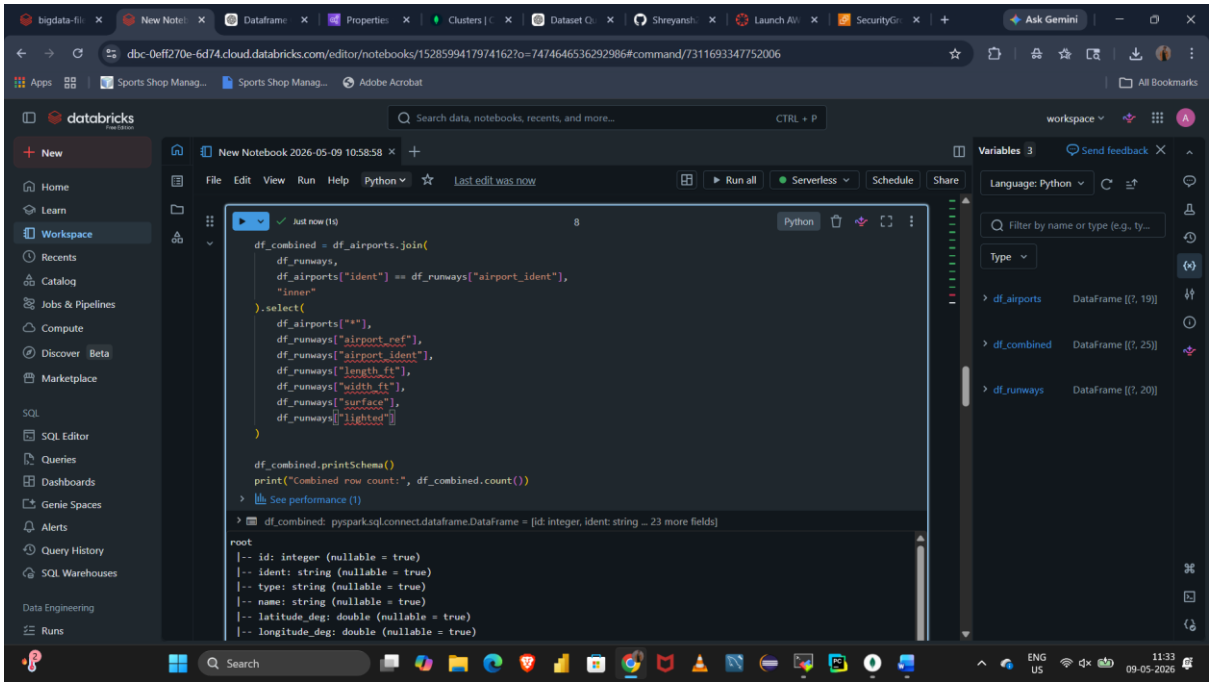
```
df_airports.show(10)
df_runways.show(10)
```

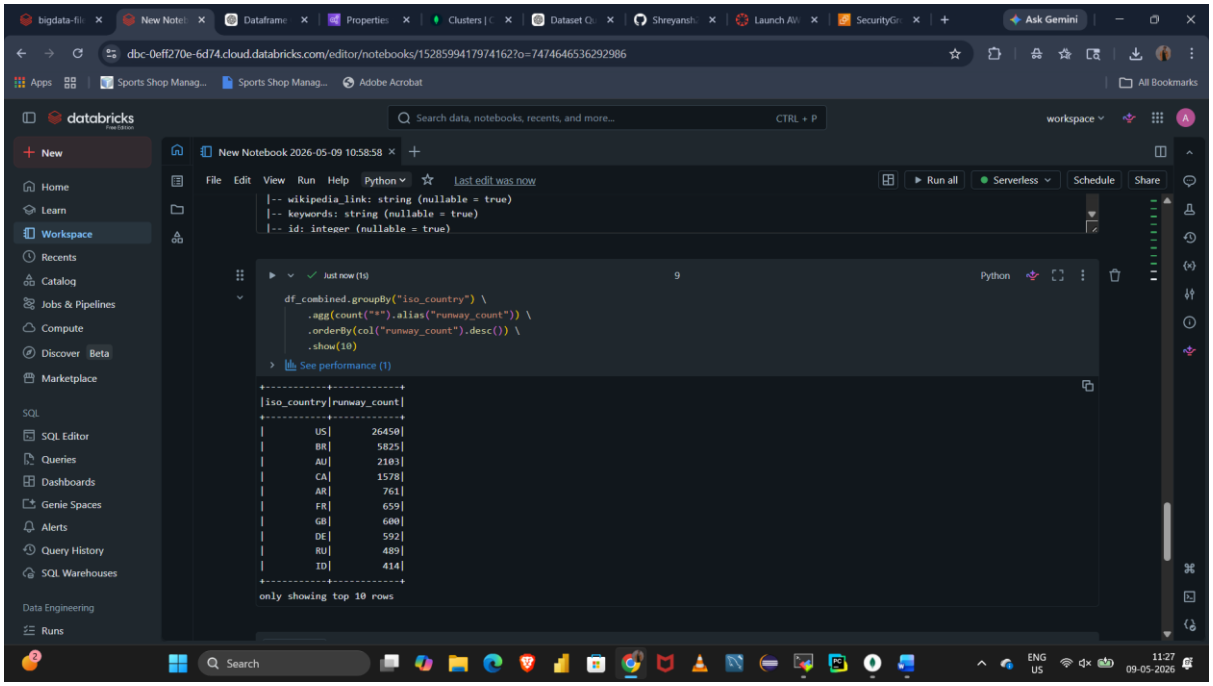
See performance (2)

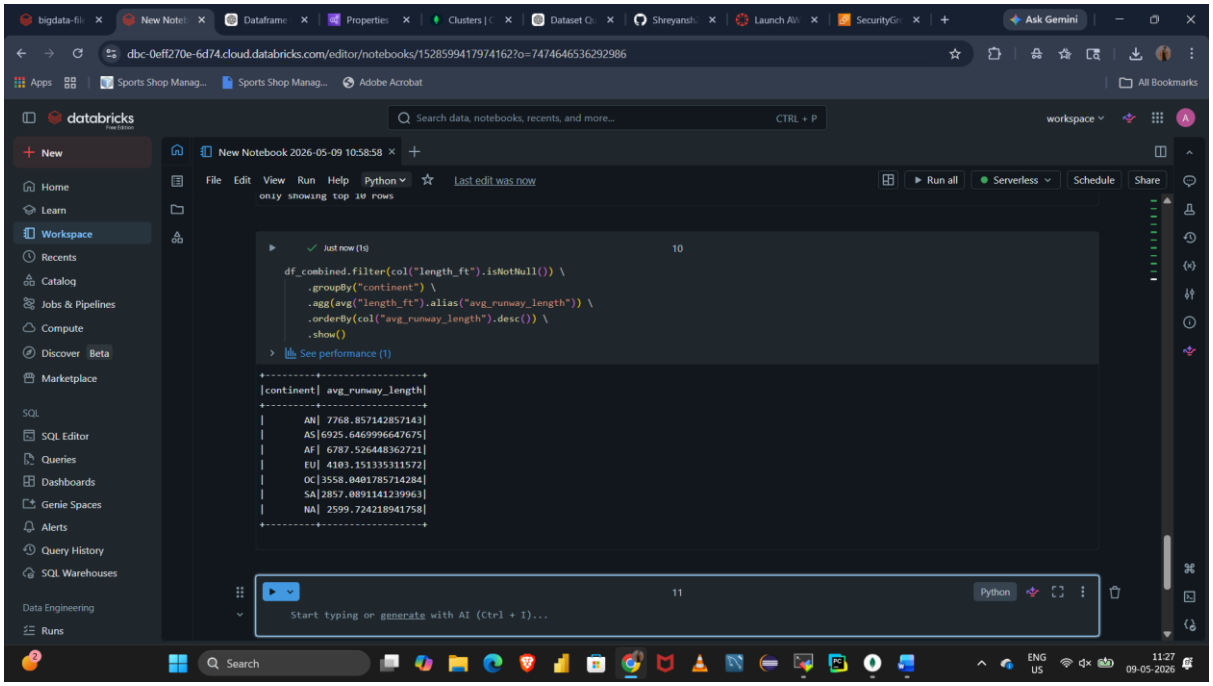
```
+------+-----+-------------+--------------------+------------------+------------------+------------+---------+-----------+----------+-------------
----------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|    id|ident|         type|                name|     latitude_deg|    longitude_deg|elevation_ft|continent|iso_country|iso_region|municipality|
scheduled_service|icao_code|iata_code|gps_code|local_code|          home_link|     wikipedia_link|keywords|
+------+-----+-------------+--------------------+------------------+------------------+------------+---------+-----------+----------+-------------
----------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
| 6523| 00A|     heliport|   Total RF Heliport|        40.070985|        -74.933689|          11|       NA|         US|     US-PA|    Bensalem|
no|     NULL|     NULL|     K00A|       00A|https://www.pennd...|                NULL|     NULL|
|323361| 00AA|small_airport|Aero B Ranch Airport|        38.704022|       -101.473911|        3435|       NA|         US|     US-KS|       Leoti|
no|     NULL|     NULL|     00AA|      00AA|                NULL|                NULL|     NULL|
| 6524| 00AK|small_airport|        Lowell Field|        59.947733|       -151.692524|         450|       NA|         US|     US-AK|Anchor Point|
```

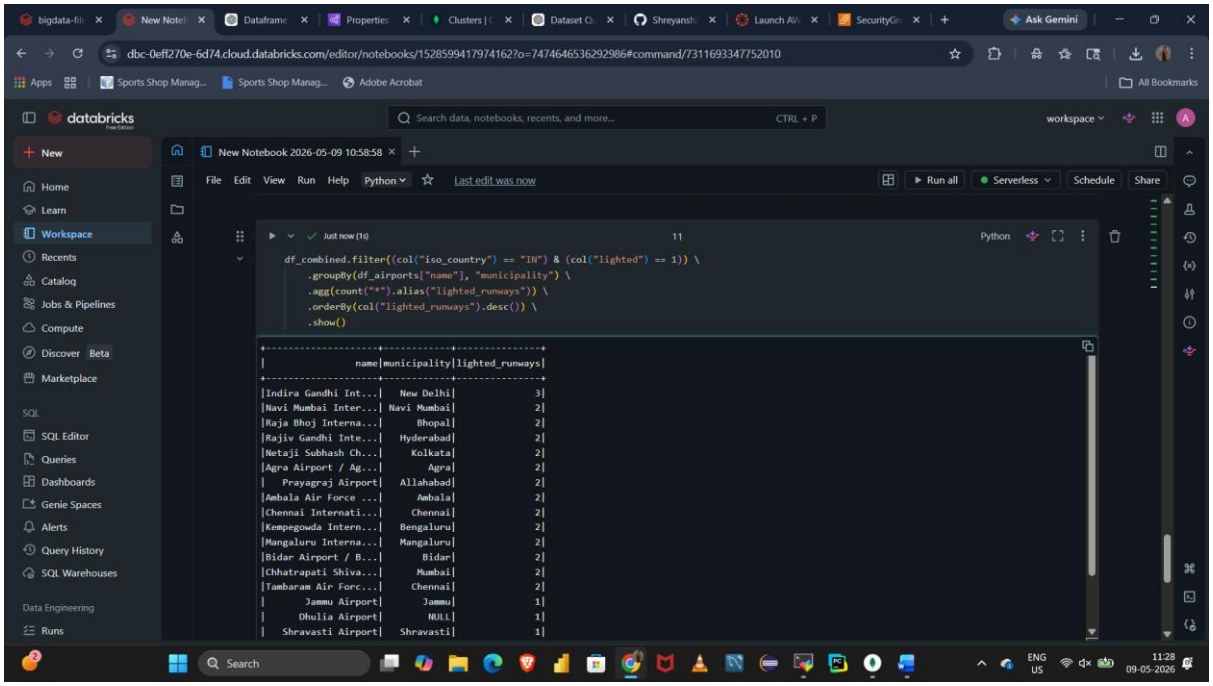

dbc-0eff270e-6d74.cloud.databricks.com/editor/notebooks/1528599417974162?o=7474646536292986
New Notebook 2026-05-09 10:58:58
Just now (1s)
6
df_airports.printSchema()
print("Airports row count:", df_airports.count())
See performance (1)
|-- id: integer (nullable = true)
|-- ident: string (nullable = true)
|-- type: string (nullable = true)
|-- name: string (nullable = true)
|-- latitude_deg: double (nullable = true)
|-- longitude_deg: double (nullable = true)
|-- elevation_ft: integer (nullable = true)
|-- continent: string (nullable = true)
|-- iso_country: string (nullable = true)
|-- iso_region: string (nullable = true)
|-- municipality: string (nullable = true)
|-- scheduled_service: string (nullable = true)
|-- icao_code: string (nullable = true)
|-- iata_code: string (nullable = true)
|-- gps_code: string (nullable = true)
|-- local_code: string (nullable = true)
|-- home_link: string (nullable = true)
|-- wikipedia_link: string (nullable = true)
|-- keywords: string (nullable = true)
Airports row count: 85324

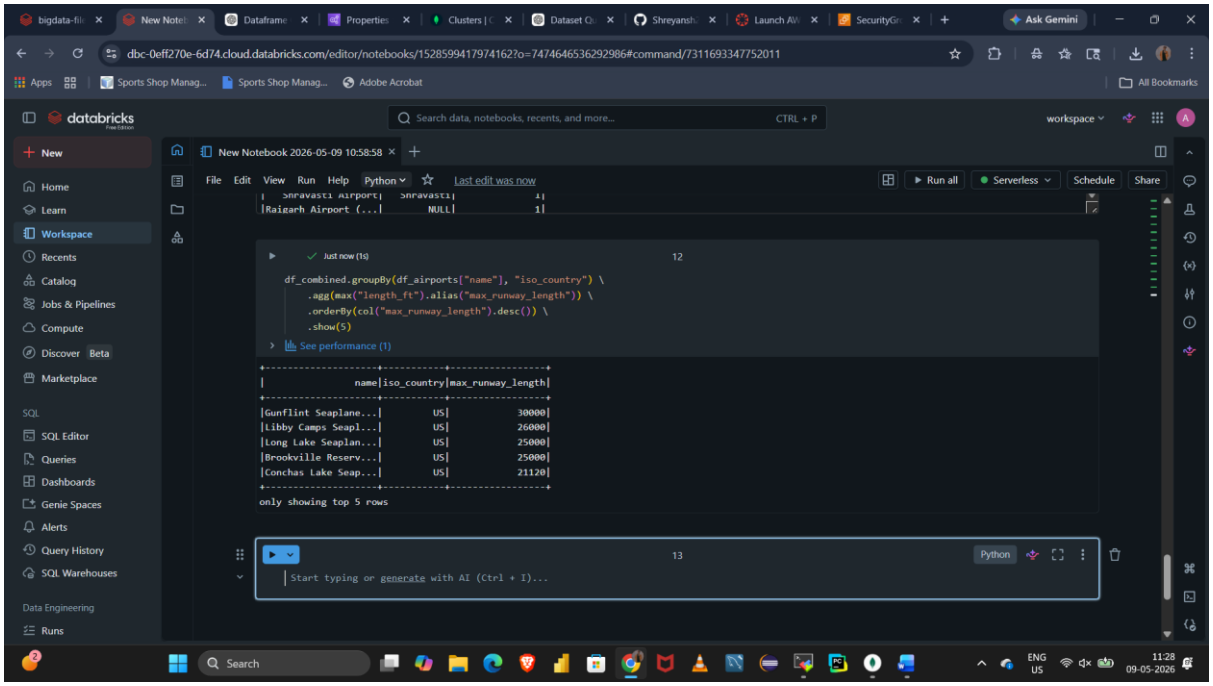

dbc-0eff270e-6d74.cloud.databricks.com/editor/notebooks/1528599417974162?o=7474646536292986#command/7311693347752006
New Notebook 2026-05-09 10:58:58
Just now (1s)
7
print("Runways row count:", df_runways.count())
|-- airport_ref: integer (nullable = true)
|-- airport_ident: string (nullable = true)
|-- length_ft: integer (nullable = true)
|-- width_ft: integer (nullable = true)
|-- surface: string (nullable = true)
|-- lighted: integer (nullable = true)
|-- closed: integer (nullable = true)
|-- le_ident: string (nullable = true)
|-- le_latitude_deg: double (nullable = true)
|-- le_longitude_deg: double (nullable = true)
|-- le_elevation_ft: integer (nullable = true)
|-- le_heading_degT: double (nullable = true)
|-- le_displaced_threshold_ft: integer (nullable = true)
|-- he_ident: string (nullable = true)
|-- he_latitude_deg: double (nullable = true)
|-- he_longitude_deg: double (nullable = true)
|-- he_elevation_ft: integer (nullable = true)
|-- he_heading_degT: double (nullable = true)
|-- he_displaced_threshold_ft: integer (nullable = true)
Runways row count: 47884
8
Start typing or generate with AI (Ctrl + I)...

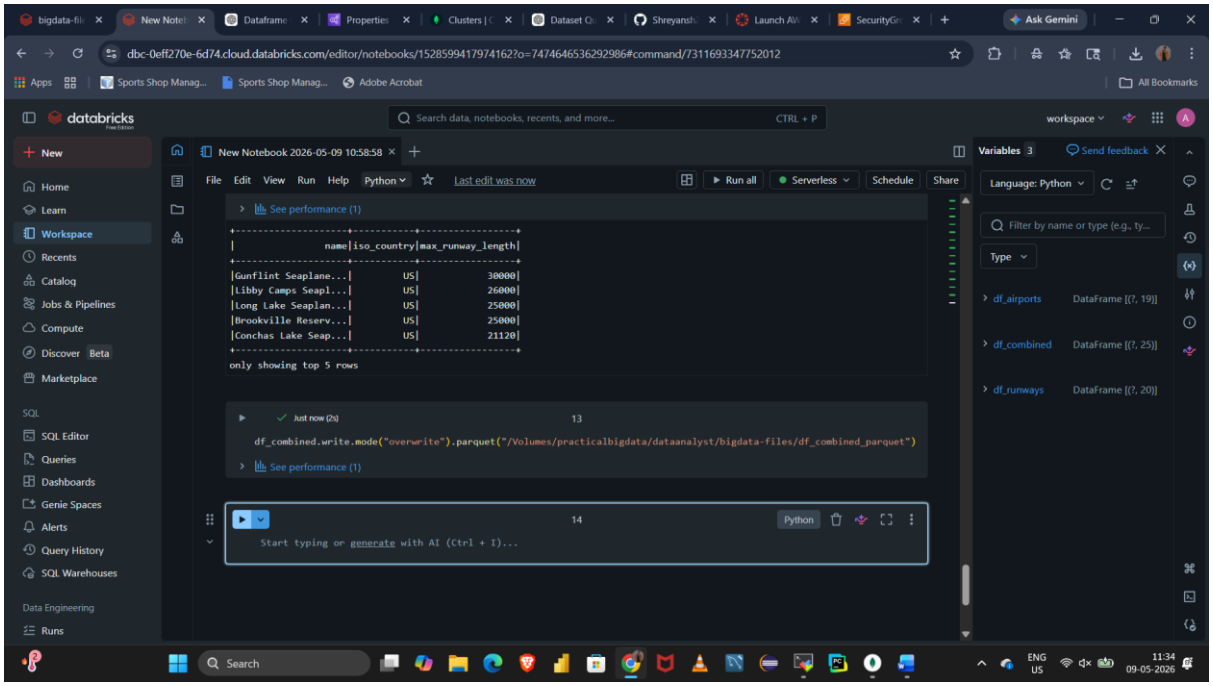

dbc-0eff270e-6d74.cloud.databricks.com/editor/notebooks/1528599417974162?o=7474646536292986#command/7311693347752006
New Notebook 2026-05-09 10:58:58
File Edit View Run Help Python Last edit was now
Run all Serverless Schedule Share
Just now (1s)
8
df_combined = df_airports.join(
df_runways,
df_airports["ident"] == df_runways["airport_ident"],
"inner"
).select(
df_airports["*"],
df_runways["airport_ref"],
df_runways["airport_ident"],
df_runways["length_ft"],
df_runways["width_ft"],
df_runways["surface"],
df_runways["lighted"]
)
df_combined.printSchema()
print("Combined row count:", df_combined.count())
See performance (1)
df_combined: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 23 more fields]
root
|-- id: integer (nullable = true)
|-- ident: string (nullable = true)
|-- type: string (nullable = true)
|-- name: string (nullable = true)
|-- latitude_deg: double (nullable = true)
|-- longitude_deg: double (nullable = true)
Variables 3
df_airports DataFrame [(?, 19)]
df_combined DataFrame [(?, 25)]
df_runways DataFrame [(?, 20)]

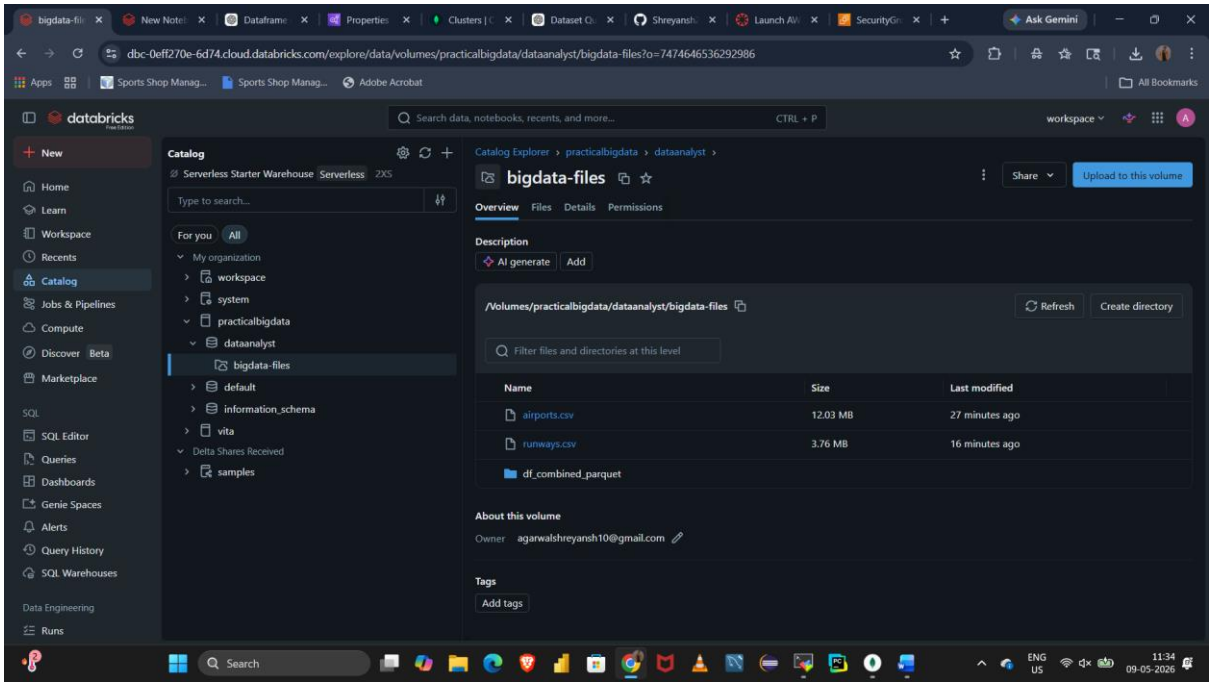

dbc-0eff270e-6d74.cloud.databricks.com/editor/notebooks/1528599417974162?o=7474646536292986
New Notebook 2026-05-09 10:58:58
|-- wikipedia_link: string (nullable = true)
|-- keywords: string (nullable = true)
|-- id: integer (nullable = true)
Just now (1s)
9
df_combined.groupBy("iso_country") \
.agg(count("*").alias("runway_count")) \
.orderBy(col("runway_count").desc()) \
.show(10)
See performance (1)
+----------+----------+
|iso_country|runway_count|
+----------+----------+
US	26450
BR	5825
AU	2103
CA	1578
AR	761
FR	659
GB	600
DE	592
RU	489
ID	414
+----------+----------+
only showing top 10 rows

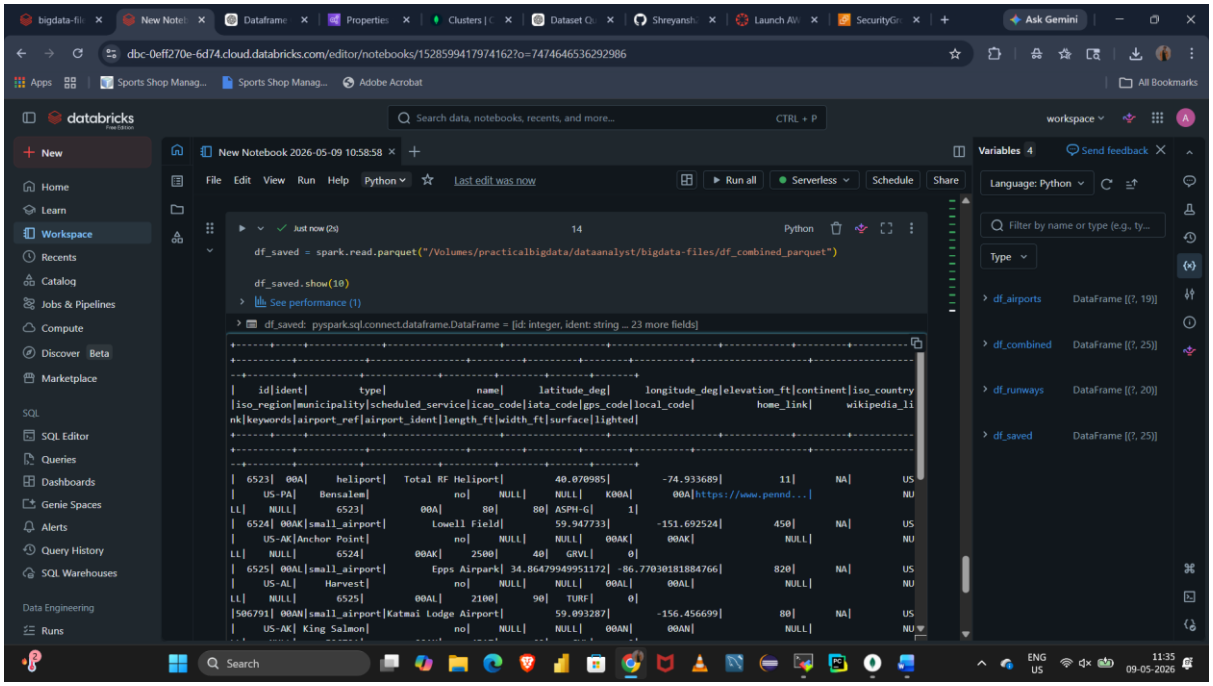

New Notebook 2026-05-09 10:58:58
only showing top 10 rows
Just now (1s)
10
df_combined.filter(col("length_ft").isNotNull()) \
.groupBy("continent") \
.agg(avg("length_ft").alias("avg_runway_length")) \
.orderBy(col("avg_runway_length").desc()) \
.show()
See performance (1)
+---------+-----------------+
|continent| avg_runway_length|
+---------+-----------------+
AN	7768.857142857143
AS	6925.6469996647675
AF	6787.526448362721
EU	4103.151335311572
OC	3558.0401785714284
SA	2857.0891141239963
NA	2599.724218941758
+---------+-----------------+
11
Start typing or generate with AI (Ctrl + I)...


New Notebook 2026-05-09 10:58:58
Just now (1s)
11
df_combined.filter((col("iso_country") == "IN") & (col("lighted") == 1)) \
.groupBy(df_airports["name"], "municipality") \
.agg(count("*").alias("lighted_runways")) \
.orderBy(col("lighted_runways").desc()) \
.show()
+--------------------+------------+---------------+
| name|municipality|lighted_runways|
+--------------------+------------+---------------+
|Indira Gandhi Int...| New Delhi| 3|
|Navi Mumbai Inter...| Navi Mumbai| 2|
|Raja Bhoj Interna...| Bhopal| 2|
|Rajiv Gandhi Inte...| Hyderabad| 2|
|Netaji Subhash Ch...| Kolkata| 2|
|Agra Airport / Ag...| Agra| 2|
| Prayagraj Airport| Allahabad| 2|
|Ambala Air Force ...| Ambala| 2|
|Chennai Internati...| Chennai| 2|
|Kempegowda Intern...| Bengaluru| 2|
|Mangaluru Interna...| Mangaluru| 2|
|Bidar Airport / B...| Bidar| 2|
|Chhatrapati Shiva...| Mumbai| 2|
|Tambaram Air Forc...| Chennai| 2|
Jammu Airport	Jammu	1
Dhulia Airport	NULL	1
Shravasti Airport	Shravasti	1


New Notebook 2026-05-09 10:58:58
Just now (1s)
12
df_combined.groupBy(df_airports["name"], "iso_country") \
.agg(max("length_ft").alias("max_runway_length")) \
.orderBy(col("max_runway_length").desc()) \
.show(5)
See performance (1)
+---+---+---+
| name|iso_country|max_runway_length|
+---+---+---+
|Gunflint Seaplane...| US| 30000|
|Libby Camps Seapl...| US| 26000|
|Long Lake Seaplan...| US| 25000|
|Brookville Reserv...| US| 25000|
|Conchas Lake Seap...| US| 21120|
+---+---+---+
only showing top 5 rows
13
Start typing or generate with AI (Ctrl + I)...


New Notebook 2026-05-09 10:58:58
See performance (1)
+---+---+---+
| name|iso_country|max_runway_length|
+---+---+---+
|Gunflint Seaplane...| US| 30000|
|Libby Camps Seapl...| US| 26000|
|Long Lake Seaplan...| US| 25000|
|Brookville Reserv...| US| 25000|
|Conchas Lake Seap...| US| 21120|
+---+---+---+
only showing top 5 rows
Just now (2s)
13
df_combined.write.mode("overwrite").parquet("/Volumes/practicalbigdata/dataanalyst/bigdata-files/df_combined_parquet")
See performance (1)
14
Start typing or generate with AI (Ctrl + I)...
Variables 3
df_airports DataFrame [(?, 19)]
df_combined DataFrame [(?, 25)]
df_runways DataFrame [(?, 20)]


dbc-0eff270e-6d74.cloud.databricks.com/explore/data/volumes/practicalbigdata/dataanalyst/bigdata-files?o=7474646536292986
databricks
Catalog
Serverless Starter Warehouse Serverless 2XS
workspace
system
practicalbigdata
dataanalyst
bigdata-files
default
information_schema
vita
Delta Shares Received
samples
Catalog Explorer › practicalbigdata › dataanalyst ›
bigdata-files
Share
Upload to this volume
Overview Files Details Permissions
Description
AI generate
Add
/Volumes/practicalbigdata/dataanalyst/bigdata-files
Refresh
Create directory
Filter files and directories at this level
Name
Size
Last modified
airports.csv
12.03 MB
27 minutes ago
runways.csv
3.76 MB
16 minutes ago
df_combined_parquet
About this volume
Owner agarwalshreyansh10@gmail.com
Tags
Add tags


dbc-0eff270e-6d74.cloud.databricks.com/editor/notebooks/1528599417974162?o=7474646536292986
New Notebook 2026-05-09 10:58:58
File Edit View Run Help Python
Last edit was now
Run all
Serverless
Schedule
Share
Just now (2s)
14
df_saved = spark.read.parquet("/Volumes/practicalbigdata/dataanalyst/bigdata-files/df_combined_parquet")
df_saved.show(10)
See performance (1)
df_saved: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 23 more fields]
Variables 4
Send feedback
Language: Python
df_airports DataFrame [(?, 19)]
df_combined DataFrame [(?, 25)]
df_runways DataFrame [(?, 20)]
df_saved DataFrame [(?, 25)]